

# AI-Powered Wholesale Order Processing for Volta Coffee



AI Automations & Agent Foundations



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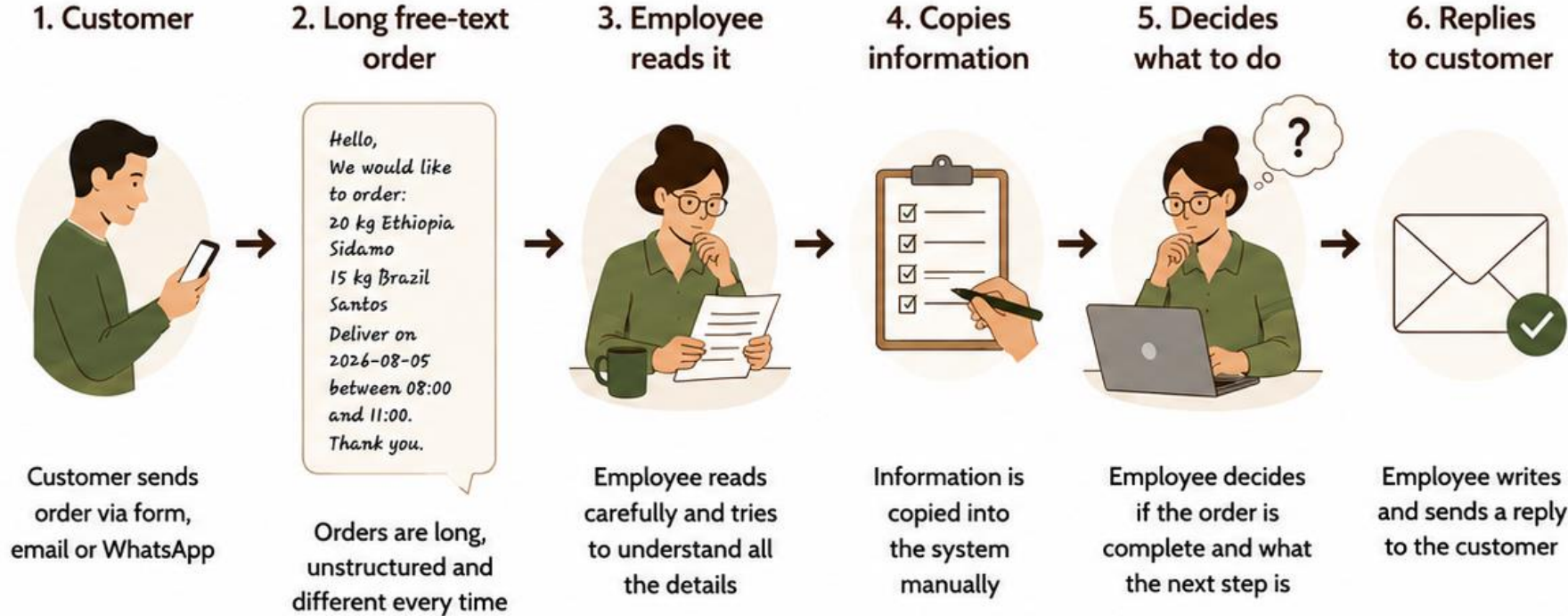
29.07.2026

  
**VOLTA**  
COFFEE ROASTERS



# The Business Problem

Wholesale orders arrive in many formats and must be processed manually.



## Problems

-  Repetitive work
-  Inconsistent information
-  Time-consuming
-  Possible mistakes



**Goal**

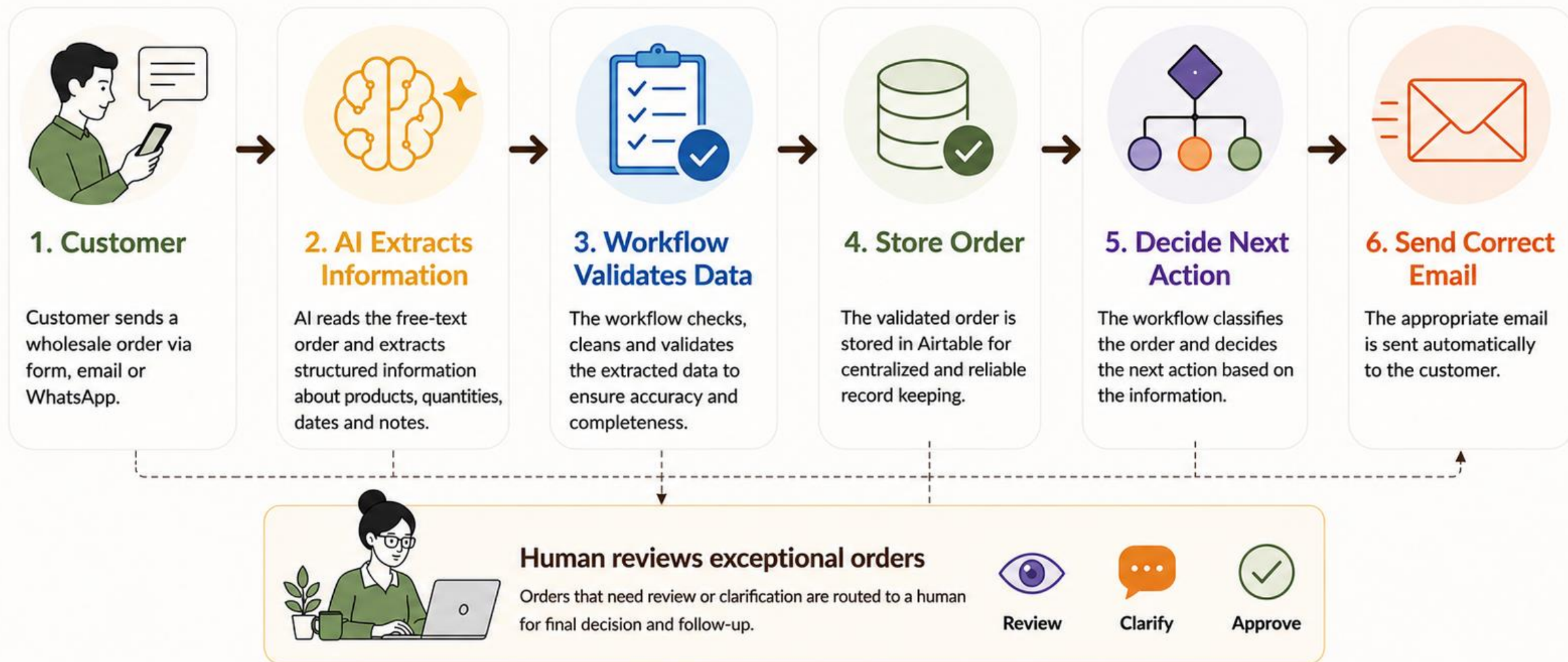
Automate repetitive tasks, not human decisions.





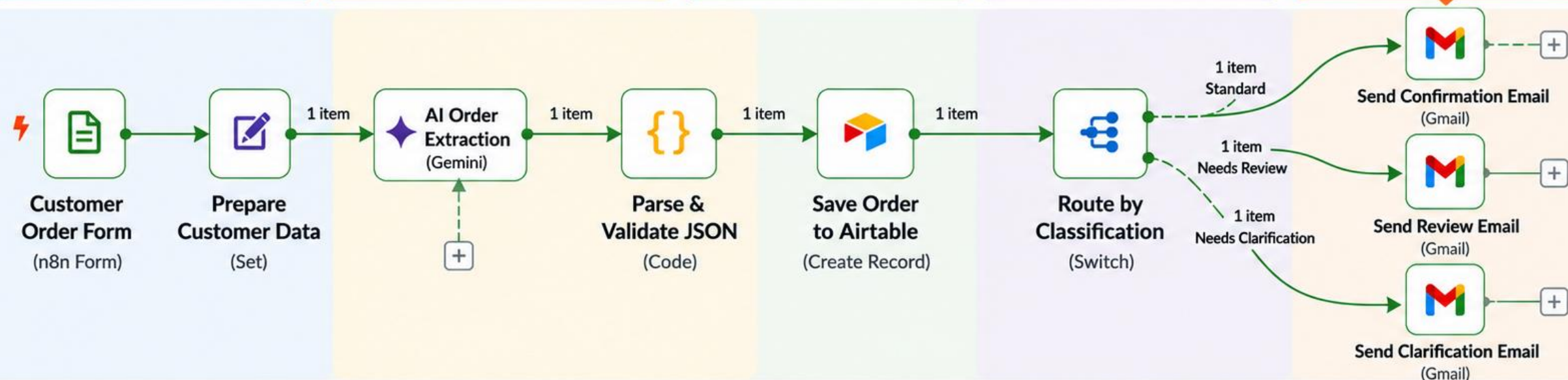
# Our Proposed Solution

An AI-powered workflow that automates the first steps and keeps humans in control when it matters.



# Workflow Architecture

An end-to-end automated workflow that processes wholesale orders using AI and routes them to the right action.



## 1. Customer Intake

Collects the raw order from the customer through an online form and prepares the data for processing.



## 2. AI Processing

AI extracts structured information from the free-text order and the workflow validates the extracted data.



## 3. Order Storage

Validated orders are stored in Airtable to keep a reliable and searchable record of all requests.



## 4. Decision Logic

The workflow classifies each order and decides the next action based on business rules.



## 5. Customer Communication

The appropriate email is sent automatically based on the classification result.



**Goal:**

Automate repetitive tasks while ensuring humans handle exceptions and important decisions.

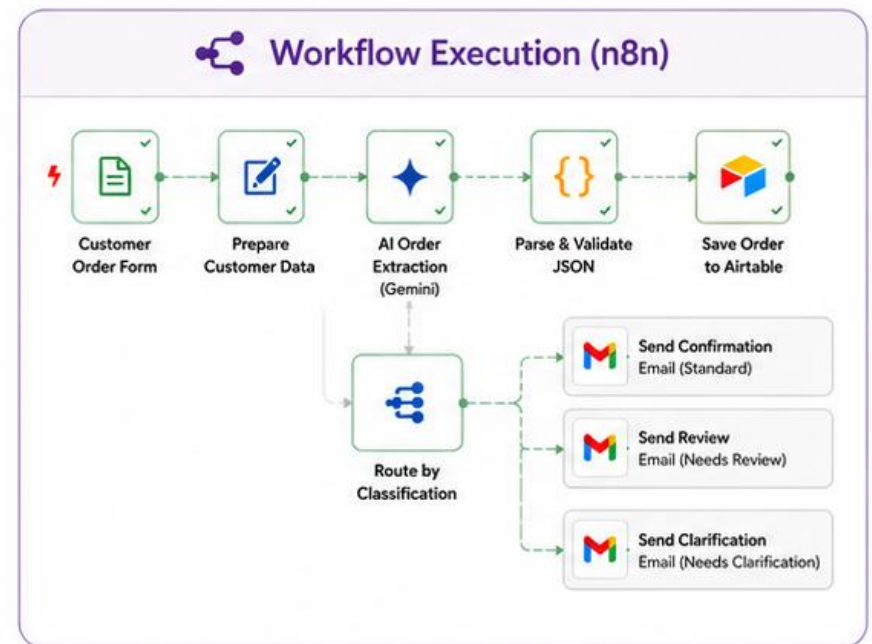






| Scenario                                                                                        | Classification      | Result                                                                              |
|-------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|
|  Standard      | Standard            |  |
|  Review        | Needs Review        |  |
|  Clarification | Needs Clarification |  |

A screenshot of an email client interface. At the top, there's a header with a Gmail logo and the text "Email (Automated Response)". Below the header is a navigation bar with icons for back, compose, reply, delete, archive, and a menu. The email subject is "Your Volta Coffee Order – Needs Review" with an "Inbox" label. The sender is "Volta Coffee Roasters" with the email address "<orders@voltacoffee.com>". The recipient is "to Sarah". The email body starts with "Hi Sarah," followed by "Thank you for your order!". Then it says "We've received your request and our team will review it shortly, especially the substitution for the House Blend." and "We will get back to you soon." It ends with "Best regards," and "Volta Coffee Roasters".



# Challenges & Improvements

During development, we faced several real-world issues.  
Each challenge helped us strengthen the workflow.

## ! CHALLENGES



### Inconsistent JSON

AI sometimes returned incomplete or malformed JSON.



### Airtable Dates

Date and time values were not always accepted by Airtable.



### Classification Normalization

Different wording from the AI caused inconsistent routing.



### Gmail Authentication

OAuth token expired or connection failed during workflow execution.



## ✓ SOLUTIONS



### Improved Prompt

Refined the prompt and added strict JSON format instructions.



### JavaScript Validation

Added validation and error handling to ensure correct data before saving.



### ISO Dates

Converted all dates to ISO 8601 format before sending to Airtable.



### Standardized Outputs

Normalized classification values and standardized email content.



Each challenge improved the reliability, accuracy, and maintainability of the workflow.  
Real engineering is solving problems, not just building features.





# What I Learned

This project helped me grow across technical, AI, and business skills.



## TECHNICAL



### n8n

Workflow automation



### APIs

Integrating external services



### Airtable

Storing and organizing data



### Gmail

Automated email communication



## AI & DATA



### Prompt Engineering

Designing effective prompts



### JSON

Structuring and parsing data



### Gemini (AI Model)

Extracting and classifying orders

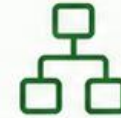


## BUSINESS



### Human-in-the-Loop

Keeping humans in control of decisions



### Workflow Design

Building clear, reliable processes



### Process Automation

Reducing manual work and improving efficiency



I learned how to combine **automation**, **AI**, and **smart workflows** to solve real business problems.

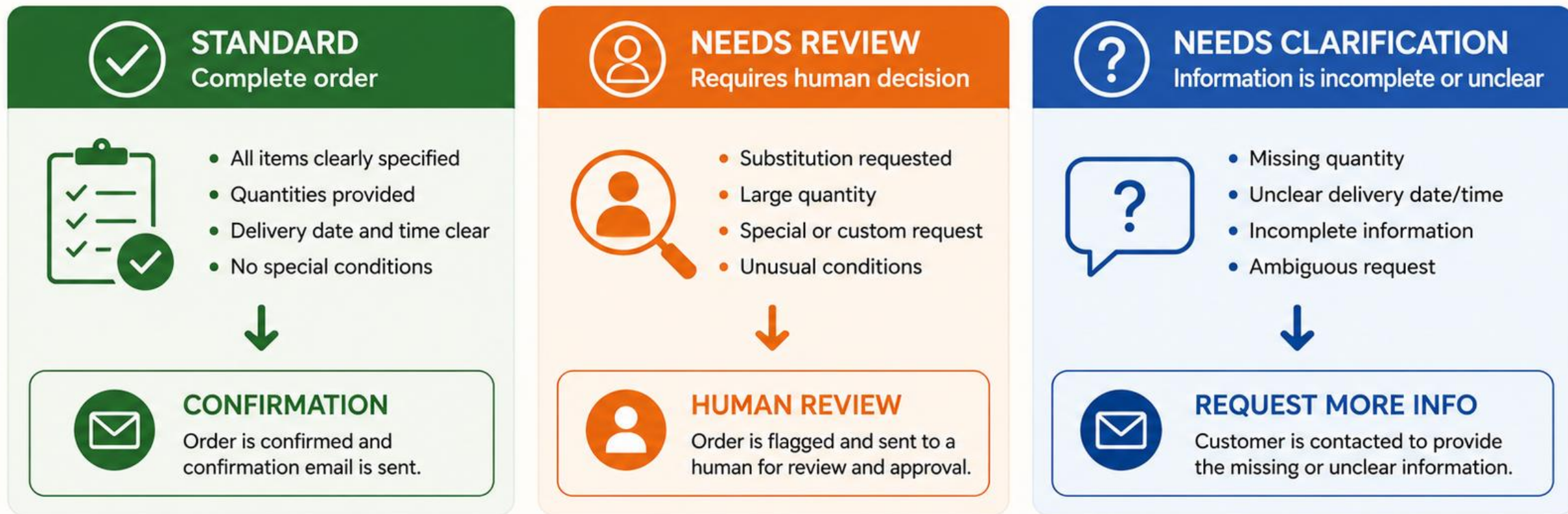






# AI Decision Making

The workflow classifies each order into one of three categories so the right action can be taken automatically.



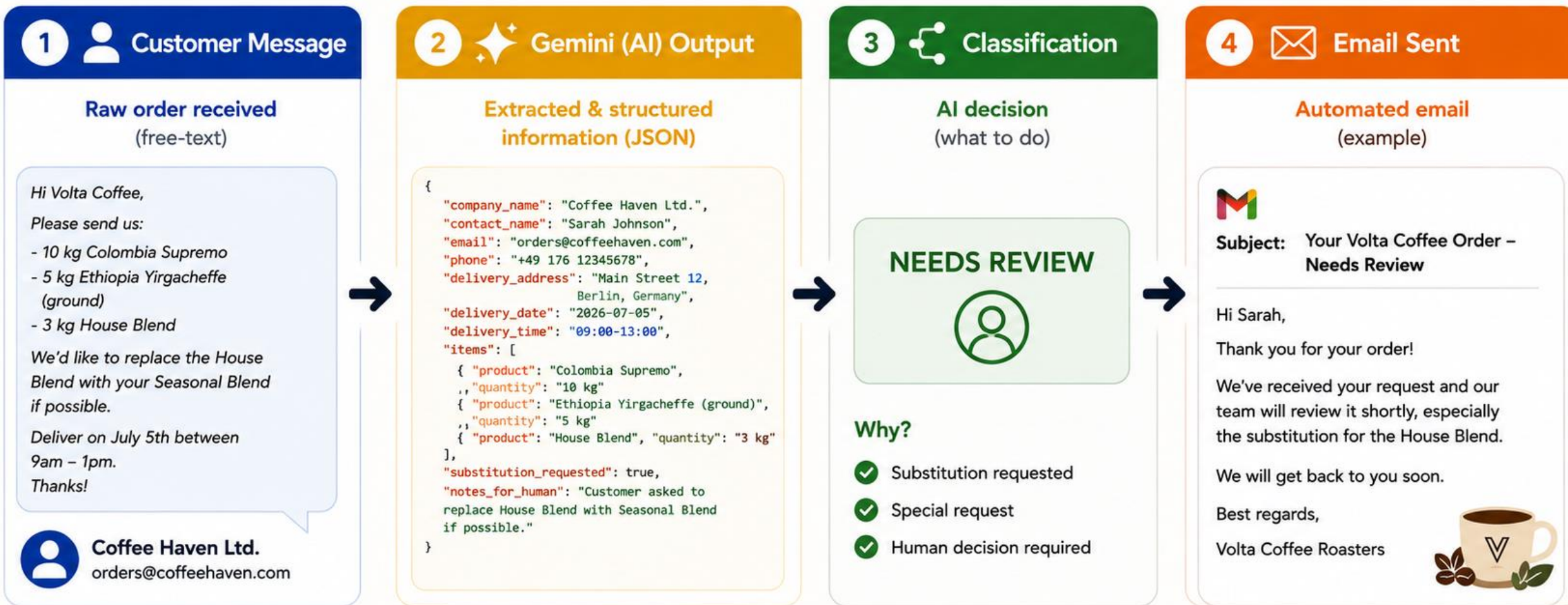
AI performs the first assessment before the workflow routes the order

Humans stay in control of important decisions.



# Example Order: From Message to Action

AI extracts the information, classifies the order, and triggers the right response.



AI understands the order and takes the **first step**.



Information is extracted, structured and stored.



Humans make the **final decision**.

